Monotone and fortuitous states in the two-flavor uplift

Progress report, 23 October 2026 (revised 25 October: closed forms)

`syk_gravity` project

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Abstract

The report of 13 October solved the BPS uplift of one-flavor $\mathcal{N} = 2$ SYK exactly. Here we do the same for the symmetrized two-flavor model of Chang, Chen, Sia and Yang, which has both monotone states (the tower $V^n|\Omega\rangle$) and fortuitous ones, and ask what in the uplift tells them apart. Adding a pair (ψ, χ) amounts to two single-mode steps, and in each step the connecting map of CV's long exact sequence can now be nonzero. We prove that the exact uplift survives with an *obstruction*. A harmonic source lifts if and only if the new couplings map it to a state with no harmonic part; when it does, the uplift is unique, orthogonally split and of minimum norm. The enlarged supercharge is $Q_0 + \psi^\dagger E + K\chi + \psi^\dagger Z\chi$, and the symmetry of the couplings gives the identity $[E, V] = K$. It makes the *current* that would dress the tower vanish. As a result the tower is, at every N, a bare superposition of old BPS states in the empty and doubly occupied slots, with weights $(N + 1 - n)/(N + 1)$ and $n/(N + 1)$. For all other states the uplift is arithmetic. Two counting laws, observed without exception, reproduce every cell of CV's channel-fidelity table: the decoder's rank fills its constrained room as far as the BPS dimension allows, and the exact-lift locus has the generic intersection dimension. The same counting says when the tower's dressing cost reaches CV's bound. This happens precisely when the charge-preserving channel stops lifting all fortuitous states of the sector. The bare BPS states are those predicted by a generic dimension count, plus the tower, which is the only state the counting does not produce. To first order, dressing costs are Euclidean-time-integrated two-point functions of three matter operators, $\psi^\dagger\chi$, $\psi^\dagger\psi^\dagger$ and $\psi^\dagger$, and these interfere between slots. Fortuitous states cancel partially; the tower cancels exactly. These two-point functions and their interference are what a chord computation of the distinction would have to produce.

Revision of 25 October. The exact uplift is now given in closed form in old-model data. The homological perturbation lemma terminates, so the new cohomology is that of the old BPS data with $D = d_1 + d_2$, where d_2 is the secondary operation $Z - EQ_0^+K + KQ_0^+E$. BPS states are $(1 + \mathcal{K})^{-1}h$ for explicitly admissible harmonic data h. Integrating out the two middle slots leaves Dyson-type equations whose self-energies are time-integrated two-point functions of E and K, together with a leak through the middle-slot cohomology. A positivity criterion guarantees that BPS states are determined by their harmonic slot data. This fails at special couplings, where we exhibit pure-dressing BPS states with no old-BPS content at all; the tower stays bare there.

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1 Introduction

1.1 Where we stood

The report of 13 October [4] solved the BPS uplift of one-flavor $\mathcal{N} = 2$ SYK exactly (`docs/derivations.md`, D6). The uplift is the passage from the BPS states of an N-mode model to those of the $(N + 1)$-mode model. Every enlarged BPS state is an old BPS state plus a dressing cloud, fixed by a linear system with an anti-Hermitian coupling. To first order, the dressing cost is the Euclidean-time-integrated two-point function of the new mode's pair operator. The report of 14 October [5] computed that two-point function from chords (D7).

In the one-flavor model every BPS state is fortuitous [2]. The natural next question is how the uplift distinguishes *monotone* states, which descend from the large-N theory, from *fortuitous* ones. The symmetrized two-flavor model of Chang, Chen, Sia and Yang (CCSY [2], eq. 3.18) is

the simplest model with both kinds. Its supercharge commutes with a bilinear V, and the tower $V^n|\Omega\rangle$ is monotone, while the rest of the BPS spectrum is fortuitous. CV [1] (§6.2) studied its uplift numerically through their channel decoders and found a sharp contrast:

- The tower lifts exactly through the charge-preserving and doubly occupied channels, with an explicit completion (CV eq. 85). Its dressing cost obeys $\kappa_{00} \leq n/(N+1-n)$, with equality "once $N \gtrsim n+3$".
- Fortuitous states lift exactly or not depending on sector and channel, in a pattern CV tabulated (their Fig. 6) but did not derive. Their costs are broad and grow with N.

CV note that "no closed form is available" for the fortuitous states.

1.2 What this report does

This report is the two-flavor analogue of the 13 October report ("Stage 1"). Its gravitational interpretation via chords ("Stage 2") is left for later. The main results are:

(i) **Exact uplift with an obstruction** (Theorem 1). Adding one mode is a mapping cone, and its connecting map need not vanish. We give the harmonic refinement of CV's long exact sequence: the explicit uplift map, the obstruction on harmonic sources, the orthogonal splitting and the canonical minimum-norm uplift. The pair uplift is two such steps (§4). A d_2 differential shows that it is *not* computed by old cohomology plus the first-order maps alone. Section 9 gives d_2 in closed form.

(ii) **The protection identity** (Proposition 1). The enlarged supercharge is $Q_0 + \psi_e^\dagger E + K\chi_f + \psi_e^\dagger Z \chi_f$. In the symmetrized model, $[E, V] = K$, a Ward identity of V.

(iii) **The tower is bare** (Proposition 2). The two slot currents that source all dressing vanish identically on the tower. Hence the tower is a superposition of old BPS states with no Hodge dressing at every N. In CV's channel language this is the "genuinely dressed" uplift (85): the two notions of dressing differ.

(iv) **Bare states are arithmetic, except the tower** (§6). A generic dimension count of the bare-state equations gives the bare BPS states exactly, in both the symmetrized and the generic model. The only excess is the tower.

(v) **CV's fidelity table derived** (§7). Two counting laws reproduce the exact-lift loci, and hence every cell of CV's Fig. 6. They are: rank $= \min(d_{\text{new}}, \text{room})$, and generic intersection inside rooms derived from the slot equations. They also give an exact criterion for when the tower's cost equals CV's bound. That criterion coincides with the point where the 00 channel stops lifting all fortuitous states.

(vi) **First order: interfering two-point functions** (§8). The first-order dressing is a quadratic form in time-integrated two-point functions of E, K and Z, with cross terms between slots. The tower is its exactly dark state. Fortuitous states interfere too, but never completely.

(vii) **The exact uplift in closed form** (§9, added 25 October). The intermediate model can be removed: everything is built from old-model data.

- *Cohomology.* The perturbation lemma gives the new cohomology as that of the old BPS data with $D = d_1 + d_2$ (Theorem 2), where d_2 is the secondary operation $\mathcal{Z} = Z - EQ_0^+K + KQ_0^+E$.

- *BPS states.* They are $(1+\mathcal{K})^{-1}h$ with explicit admissibility (Theorem 3).
- *Reduced system.* Integrating out the middle slots leaves two Dyson equations whose self-energies are time-integrated two-point functions, plus a leak through the middle cohomology (Proposition 3).
- *Uniqueness.* A criterion guarantees it, but four-slot injectivity fails at special couplings, where pure-dressing BPS states appear (§9.4).

Status of the claims. [derived] marks statements proved in the text, and [derived + verified] those also checked numerically to machine precision (Appendix A). [numerical] marks observations from exact diagonalization (five realizations at $N \leq 6$, two at $N = 7$, plus a generic-coupling control). [interpretation] marks readings that are not derivations. Table 1 summarizes.

Table 1: Main results and their status.

Result	Status	Where
Block form; $[E, K] = \{Q_0, Z\}$; slot equations	[derived + verified]	§2
Protection identity $[E, V] = K$ (symmetrized couplings only)	[derived + verified]	Prop. 1
One mode with obstruction: uplift, splitting, minimum norm	[derived + verified]	Thm. 1
Pair = two obstructed steps; dimension formula	[derived + verified]	§4
Injectivity of the four-slot harmonic map (generic; see §9.4)	[numerical]	§4
$d_2 \neq 0$: the slot complex overcounts in four sectors per N	[numerical]	§4
Tower is bare, slot weights $(N+1-n)/(N+1)$, $n/(N+1)$	[derived + verified]	Prop. 2
Bare states = generic count + tower	[numerical]	§6
Rooms of CV's decoders	[derived]	§7.1
Laws L1, L2 (rank and exact-lift locus); CV Fig. 6 reproduced and explained	[numerical]	§7
$\kappa_{00}(\text{tower}) = n/(N+1-n) \iff \ker D_{00} = 0 \iff d_{\text{new}} \leq R_{00}$	[numerical] (+ L1)	§7.4
First-order dressing = interfering two-point functions; tower exactly dark	[derived + verified]	§8
Fortuitous states interfere partially; first-order sector means within 4–6 %	[numerical]	§8
Added 25 October: closed forms		
Cohomology $= H(E_1, d_1 + d_2)$, $d_2 = P_H \mathcal{Z} P_H$; dimension formula from old data	[derived + verified]	Thm. 2
BPS states $= (1+\mathcal{K})^{-1}h$, admissibility (a)–(c), slot weights, decoder pencils	[derived + verified]	Thm. 3, Prop. 5
Reduced system: self-energies $EGE^\dagger + KGK^\dagger$, transfer $\mathcal{Z}$, factorized leak	[derived + verified]	Prop. 3
Uniqueness criterion $\rho < 2$	[derived]	Prop. 4
Four-slot injectivity fails at special couplings (pure-dressing states)	[numerical]	Table 7
Self-energies alone recover 56–92 % of the first-order error in the bare fraction	[numerical]	§9

2 The model and the pair uplift

2.1 The symmetrized two-flavor model

There are N complex fermions of each of two flavors, ψ_i and χ_i ($i = 1, \ldots, N$). We write creation operators with explicit daggers. CCSY's supercharge (eq. 3.18; CV eq. 82) is

$$Q = \sum_{i,j,k=1}^{N} C_{ijk}\, \psi_i^\dagger \psi_j^\dagger \chi_k, \qquad C_{ijk} = C_{ikj}, \tag{2.1}$$

with $H = \{Q, Q^\dagger\}$. We take $C = (G + G^{T_{jk}})/2$ with G i.i.d. standard complex Gaussian; only ratios of couplings matter below. Q has charge $(+2, -1)$ under (N_ψ, N_χ), and H conserves both. We label sectors by $(a, b) = (N_\psi, N_\chi)$.

The symmetry in the last two indices makes Q commute with

$$V = \sum_i \psi_i^\dagger \chi_i^\dagger, \qquad [Q, V] = 0 \qquad \text{(CCSY 3.19)}. \tag{2.2}$$

The tower $V^n|\Omega\rangle$ ($n = 0, \ldots, N$) lies in sector (n, n) and has norm $\|V^n|\Omega\rangle\|^2 = n!\, N!/(N-n)!$ (CV 86). It is Q-closed by (2.2). It is $Q^\dagger$-closed because V, $V^\dagger$ and $N_V = (N_\psi + N_\chi - N)/2$ form an $\mathfrak{su}(2)$ for which the tower is the spin-$N/2$ multiplet of the vacuum (CCSY 3.22–3.23). CCSY show that it is monotone. Following CV, *fortuitous* means the orthogonal complement of the tower in each sector's BPS space.

As a control we use the *generic* model of Heydeman, Turiaci and Zhao [3]: i.i.d. couplings with no symmetry. There V does not commute with Q and there is no tower.

BPS dimensions are coupling-independent at these sizes. [numerical] In every sector, for all $N' \le 8$, the BPS dimensions of the symmetrized and generic models coincide in all realizations. For example, sector $(4, 4)$ at $N' = 8$ has dimension 1820 in both. The symmetrization therefore does not create cohomology at these sizes; it makes one state per diagonal sector special. This bears on CCSY's footnote 18, which suggests that tower states "could easily be lifted" by non-symmetric perturbations. Here no BPS state is lifted. One state is replaced by a generic one.

2.2 Adding a pair: the block form

Passing from N to $N+1$ adds the pair (ψ_e, χ_f), both carrying the coupling index $N+1$. The occupations of the new pair define four *slots* $T = (t_\psi, t_\chi) \in \{00, 10, 01, 11\}$, CV's four channels. We use the operator convention

$$\Psi = \Psi_{00} + \psi_e^\dagger \Psi_{10} + \chi_f^\dagger \Psi_{01} + \psi_e^\dagger \chi_f^\dagger \Psi_{11}, \tag{2.3}$$

where Ψ_T is a state of the old model in sector $(A - t_\psi, B - t_\chi)$ when Ψ is in sector (A, B). Collecting the terms of the enlarged supercharge that contain ψ_{N+1}, χ_{N+1} or both gives

$$Q_{N+1} = Q_0 + \psi_e^\dagger E + K \chi_f + \psi_e^\dagger Z \chi_f, \tag{2.4}$$

with three operators on the old modes:

$$E = \sum_{i,k} (C_{eik} - C_{iek})\, \psi_i^\dagger \chi_k, \qquad K = \sum_{i<j} (C_{ijf} - C_{jif})\, \psi_i^\dagger \psi_j^\dagger, \qquad Z = \sum_i (C_{eif} - C_{ief})\, \psi_i^\dagger, \tag{2.5}$$

where $e = f = N + 1$ as a coupling index. E converts a χ into a ψ (even, charge $(+1, -1)$). K creates a ψ pair (even, $(+2, 0)$). Z creates a single ψ (odd, $(+1, 0)$). Expanding $Q_{N+1}^2 = 0$ in powers of $\psi_e^\dagger$ and χ_f gives

$$[Q_0, E] = 0, \qquad [Q_0, K] = 0, \qquad [E, K] = \{Q_0, Z\}. \tag{2.6}$$

[derived + verified] Equation (2.4) holds as an operator identity, and (2.6) to machine precision, in both models.

Proposition 1 (Protection identity)**.** *[derived + verified] In the symmetrized model,*

$$[E, V_N] = K, \qquad [K, V_N] = 0. \tag{2.7}$$

For generic couplings the first identity fails.

Proof. Write $V_{N+1} = V_N + \psi_e^\dagger \chi_f^\dagger$, an even operator that commutes with V_N. The term of $[Q_{N+1}, V_{N+1}] = 0$ proportional to $\psi_e^\dagger$ alone is $\psi_e^\dagger([E, V_N] - K)$. The $-K$ comes from $[K\chi_f, \psi_e^\dagger \chi_f^\dagger] = -\psi_e^\dagger K$. The term proportional to χ_f is $[K, V_N]\chi_f$, and $[K, V_N] = 0$ holds trivially because both contain only creation operators. Directly, $[\psi_i^\dagger \chi_k, V] = -\psi_i^\dagger \psi_k^\dagger$, so $[E, V] = -\sum_{i<k}(E_{ik} - E_{ki})\psi_i^\dagger \psi_k^\dagger$. The identity is therefore $K_{ik} = E_{ki} - E_{ik}$, which reduces to $C_{N+1,i,k} = C_{N+1,k,i}$: the symmetry of the couplings in their last two indices, at the new index. Numerically the residual is 10^{-16} in the symmetrized model and $O(1)$ for generic couplings. □

2.3 The slot equations

Inserting (2.3) into $Q_{N+1}\Psi = 0$ and $Q_{N+1}^\dagger \Psi = 0$ and separating slots gives eight equations (Table 2). Two combinations play a central role. We call them the *slot currents*:

$$J_{10} := E\Psi_{00} - K\Psi_{11} + Z\Psi_{01}, \qquad J_{01} := K^\dagger \Psi_{00} + E^\dagger \Psi_{11} + Z^\dagger \Psi_{10}. \tag{2.8}$$

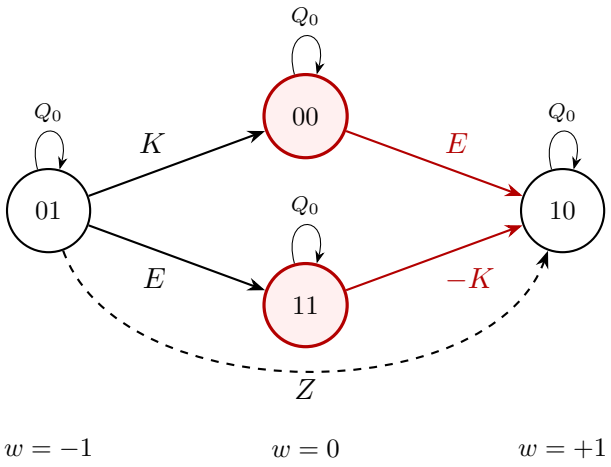
Table 2: The slot equations (S), i.e. the BPS conditions $Q_{N+1}\Psi = Q_{N+1}^\dagger \Psi = 0$ in the convention (2.3). [derived + verified] on every BPS state at $N = 4$ in both models (`tests/test_two_flavor.py`).

slot	from $Q_{N+1}\Psi = 0$	from $Q_{N+1}^\dagger \Psi = 0$
00	$Q_0\Psi_{00} = -K\Psi_{01}$	$Q_0^\dagger \Psi_{00} = -E^\dagger \Psi_{10}$
10	$Q_0\Psi_{10} = J_{10}$	$Q_0^\dagger \Psi_{10} = 0$
01	$Q_0\Psi_{01} = 0$	$Q_0^\dagger \Psi_{01} = J_{01}$
11	$Q_0\Psi_{11} = -E\Psi_{01}$	$Q_0^\dagger \Psi_{11} = K^\dagger \Psi_{10}$

Three immediate consequences:

- Ψ_{01} is closed and Ψ_{10} is coclosed.
- Write each component in the old model's Hodge decomposition, harmonic ⊕ exact ⊕ coexact. The non-harmonic parts of Ψ_{00} and Ψ_{11} are fixed by Ψ_{01} and Ψ_{10}, for example $\text{coexact}(\Psi_{00}) = -Q_0^+ K\Psi_{01}$ and $\text{exact}(\Psi_{00}) = -(Q_0^\dagger)^+ E^\dagger \Psi_{10}$.
- The coexact part of Ψ_{10} is $Q_0^+ J_{10}$ and the exact part of Ψ_{01} is $(Q_0^\dagger)^+ J_{01}$. All *dressing* of a BPS state, meaning its non-harmonic slot content, is therefore sourced by $K\Psi_{01}$, $E\Psi_{01}$, $E^\dagger \Psi_{10}$, $K^\dagger \Psi_{10}$ and the two currents.

Figure 1 shows the structure. Q_{N+1} raises $w := t_\psi - t_\chi$ by 0 (Q_0), 1 (E, K) or 2 (Z). The slots are therefore filtered as $01 \to \{00, 11\} \to 10$.



$J_{10} = E\Psi_{00} - K\Psi_{11} + Z\Psi_{01}$

tower: $EV^n|\Omega\rangle = nKV^{n-1}|\Omega\rangle$

$\Rightarrow J_{10} = 0$ (and $J_{01} = 0$)

Figure 1: The four slots of the pair uplift, ordered by $w = t_\psi - t_\chi$, and the parts of $Q_{N+1} = Q_0 + \psi_e^\dagger E + K\chi_f + \psi_e^\dagger Z\chi_f$ connecting them; Q_0 acts within each slot. The adjoint flows into slot 01 form the current J_{01}. All dressing is sourced by these arrows. The tower (red) occupies slots 00 and 11 only. Its two arrows into slot 10 cancel by the protection identity $[E, V] = K$, and the adjoint arrows into 01 cancel as well, so the tower needs no dressing at all.

Table 3: Notation.

Symbol	Meaning
(a, b), (A, B)	old and new sectors (N_ψ, N_χ)
$T = (t_\psi, t_\chi)$; Ψ_T	slot (channel) of the new pair; slot component, a state of the old model
Q_0, H_0, $G = H_0^+$	old supercharge, Hamiltonian, Green's operator $\int_0^\infty d\tau\, e^{-\tau H_0}$ (non-harmonic part)
$\mathsf{B}(a, b)$, $b(a, b)$, P_H	old BPS space, its dimension, its projector
E, K, Z	couplings to the new pair, eq. (2.5)
J_{10}, J_{01}	slot currents, eq. (2.8)
d_{new}	dimension of the new BPS space in the target sector
D_T, U_T, κ_T	CV's decoder $\Pi_T\|_{\mathsf{B}_{N+1}}$, minimum fidelity over the fortuitous subspace, dressing cost
R_T, A_T	constrained room containing $\operatorname{im} D_T$; exact-lift locus $\mathsf{B} \cap \operatorname{im} D_T$
bare; bare fraction μ	all slot components old-harmonic; $\sum_T \|P_H \Psi_T\|^2 / \|\Psi\|^2$

3 Adding one mode with an obstruction

The one-flavor uplift (D6, at half filling) rested on a dimension fact: the image of a harmonic source under the new couplings lands in a charge sector without cohomology, so it is automatically exact. In the two-flavor model this fails. The targets of E and K lie inside the window where the old model has BPS states. The following theorem is the replacement.

Theorem 1 (One added mode with an obstruction)**.** *[derived + verified] Let Q_0 be nilpotent on a finite-dimensional graded Hilbert space $\mathcal{H}_o$. Add one fermionic mode μ, with $Q = Q_0 + \mu^\dagger E$, where E is even and $[Q_0, E] = 0$ ("creation type"). Write $\Psi = a + \mu^\dagger b$ with $a, b \in \mathcal{H}_o$, and Hodge-decompose $a = h + x$ and $b = g + z$ (h, g harmonic, x exact, z coexact).*

(a) Ψ is BPS if and only if $Q_0 a = 0$, $Q_0^\dagger b = 0$, $Q_0 b = Ea$ and $Q_0^\dagger a = -E^\dagger b$.

(b) For given harmonic sources (h, g) a BPS state exists if and only if

$$P_H E\, h = 0 \qquad \textit{and} \qquad P_H E^\dagger g = 0. \tag{3.1}$$

It is then unique, with $z = Q_0^+ E(h + x)$ and $x = -(Q_0^\dagger)^+ E^\dagger(g + z)$.

(c) Hence $\mathsf{B} \cong \ker(P_H E P_H|_{\mathsf{B}_a}) \oplus \ker(P_H E^\dagger P_H|_{\mathsf{B}_b})$. The BPS space splits orthogonally as $\mathsf{B} = \mathsf{B}_h \oplus \mathsf{B}_g$ (no harmonic part in the occupied, resp. empty, slot), and $\Psi(h, 0)$ is the unique minimum-norm BPS state whose empty-slot harmonic part is h.

For an annihilation-type mode, $Q = Q_0 + K\mu$, the roles of the slots are exchanged. Then $a = h + z$ and $b = g + x$, and the obstructions are $P_H K g = 0$ and $P_H K^\dagger h = 0$.

Proof. (a) $Q\Psi = Q_0 a + \mu^\dagger(Ea - Q_0 b)$ and $Q^\dagger\Psi = Q_0^\dagger a + E^\dagger b - \mu^\dagger Q_0^\dagger b$.

(b) The equation $Q_0 b = Ea$ requires Ea to be exact. Now $Ex = EQ_0 w = Q_0 E w$ is exact. Eh is closed, because $Q_0 Eh = EQ_0 h = 0$, and closed forms are harmonic $\oplus$ exact. So Ea is exact if and only if $P_H Eh = 0$. Then $Q_0 z = Ea$ has the unique coexact solution $z = Q_0^+ Ea$. The coclosed equation is the mirror image. The remaining linear system for (x, z), given (h, g), is the one of D6 §5. Its proof uses only $[Q_0, E] = 0$ and the exactness of Ea, so it carries over verbatim. The coupling is anti-Hermitian, so $1 + \mathbb{L}$ is invertible and the solution is unique.

(c) The orthogonality of $\Psi(h, 0)$ and $\Psi(0, g)$, and the minimum-norm property, follow as in D6 (Theorem 2 and Corollary 1 of [4]). □

Remark (Cohomological reading). Q is the mapping cone of $E : (\mathcal{H}_o, Q_0) \to (\mathcal{H}_o, Q_0)$. Its long exact sequence is CV's eq. (24), with connecting map $\delta = [E]$, and the dimension count in (c) is CV's eq. (26). In the half-filling sector of the one-flavor model used in D6, $\delta = 0$ because its target lies outside the fortuity window. (In other one-flavor sectors δ can be nonzero; CV §4.2 show that the incoming and outgoing maps cannot both be.) Theorem 1 is the harmonic, metric refinement for $\delta \neq 0$. It gives the explicit uplift map, the obstruction (3.1) on harmonic sources, the orthogonal splitting and the canonical uplift. CV §4.3 emphasized that these require "the Hilbert-space metric and the coupling-dependent supercharge of the enlarged theory."

[derived + verified] We checked Theorem 1 at $N = 4, 5$ in both models by solving the BPS conditions for the non-harmonic parts with prescribed harmonic sources:

- admissible sources give a BPS state with residual $\leq 10^{-12}$, with no rank deficit;
- obstructed sources give residual ≥ 1;
- the splitting holds to 10^{-13}.

4 The pair uplift

Two obstructed steps. We add ψ_e first and χ_f second.

- Step 1 is of creation type with the operator E. It produces an intermediate model M, in which χ_f is frozen empty and $Q_M = Q_0 + \psi_e^\dagger E$.
- Step 2 is of annihilation type over M, with $K' = K + \psi_e^\dagger Z$. The required $[Q_M, K'] = 0$ is equation (2.6).

Theorem 1 applied twice gives the BPS space of the enlarged model explicitly, and in particular

$$\dim \mathsf{B}_M(A,B) = \dim\ker\left(P_H E P_H\big|_{\mathsf{B}(A,B)}\right) + \dim\ker\left(P_H E^\dagger P_H\big|_{\mathsf{B}(A-1,B)}\right), \tag{4.1}$$

$$\dim \mathsf{B}_{N+1}(A,B) = \dim\ker\left(P_H K'^\dagger P_H\big|_{\mathsf{B}_M(A,B)}\right) + \dim\ker\left(P_H K' P_H\big|_{\mathsf{B}_M(A,B-1)}\right). \tag{4.2}$$

[derived + verified] There are no mismatches in any sector for $N = 2$–7: 950 symmetrized and 570 generic sector records for $N \le 6$, plus 81 sector records (seed 1) at $N = 7$.

The four-slot harmonic map. [numerical] A BPS state whose four slot components all have vanishing old-harmonic part is zero. Equivalently, $\mathsf{B}_{N+1}(A,B)$ embeds in $\bigoplus_T \mathsf{B}(\text{slot } T)$. We find no rank deficit for any sector at $N = 2$–7 in either model, and every BPS state is reconstructed from its four harmonic slot parts to 10^{-14}. We have no proof. The obstacle is that the old-harmonic part of Ψ_{00} is $h_{00} + P_H E^\dagger d$, where d comes from the M-coexact dressing of step 2, rather than the step-1 source h_{00} itself. Injectivity is not robust, though. The smallest eigenvalue of the operator (4.3) below falls to 1.8×10^{-6} (seed 1) and 5×10^{-5} (generic) in sectors $(3,3)$ and $(5,5)$ at $N = 7 \to 8$. *Update (§9.4):* this happens where the old model has a near-zero mode. Injectivity is generic, but it fails exactly on a codimension-two set of couplings, where pure-dressing BPS states exist. Where it holds, injectivity makes the following operator positive definite on B_{N+1}:

$$\mathcal{G} = \sum_T S_T^\dagger\, P_H\, S_T \qquad (S_T :\ \text{slot-}T\ \text{component}). \tag{4.3}$$

Its eigenvalue on a state is the *bare fraction* $\mu \in (0,1]$: the share of the norm carried by old BPS states in the four slots. A state is *bare* if $\mu = 1$, i.e. if its Hodge dressing vanishes. In the one-flavor model, at first order, $\mu = 1/(1+\sigma^2)$ coincides with the decoder eigenvalue λ on the h family and with $1-\lambda$ on the g family. $\mathcal{G}$ is the slot-symmetric version of that quantity, and it is the natural object when there are four slots.

Old cohomology plus $[E],[K]$ is not enough. Because the pair has three filtration levels, one may try to compute B_{N+1} from the *slot complex* of old BPS data,

$$\mathsf{B}(01) \xrightarrow{([K],[E])} \mathsf{B}(00) \oplus \mathsf{B}(11) \xrightarrow{[E]\,h_{00} - [K]\,h_{11}} \mathsf{B}(10). \tag{4.4}$$

This is the E_1 page of the spectral sequence of the w filtration, and $d_1^2 = [\{Q_0, Z\}] = 0$. Its cohomology E_2 bounds $\dim \mathsf{B}_{N+1}$ from above. [numerical] The bound is saturated in all sectors but four per N, identically in both models (Table 4). There a differential $d_2 \sim [Z - EQ_0^+K + KQ_0^+E]$ pairs the $w = -1$ classes of sector (A,B) with the $w = +1$ classes of $(A+2, B-1)$. So the pair uplift is *not* determined by the old BPS data and the first-order maps alone. The two-step formula (4.2) is exact and accounts for d_2.

5 The tower: protection by current cancellation

Proposition 2 (The tower is bare)**.** *[derived + verified] In the symmetrized model, the enlarged tower state $V_{N+1}^n|\Omega\rangle$ has slot components*

$$\Psi_{00} = V^n|\Omega\rangle, \qquad \Psi_{11} = n\,V^{n-1}|\Omega\rangle, \qquad \Psi_{10} = \Psi_{01} = 0, \tag{5.1}$$

all of them old BPS states. Both slot currents vanish, and the tower has bare fraction $\mu = 1$ at every N. Its slot weights are $(N+1-n)/(N+1)$ in 00 and $n/(N+1)$ in 11.

Table 4: The four sectors per N where the slot complex (4.4) overcounts: $\dim E_2 \to \dim \mathsf{B}_{N+1}$. In every case E_2 sits entirely at $w=-1$ (sectors with $A<B$) or at $w=+1$ (sectors with $A>B$). d_2 maps sector (A,B) to $(A+2,B-1)$ with the maximal rank: 9, 78, 91 at $N=5,6,7$, for example. Identical in all realizations of both models.

$N \to N+1$	sector: $\dim E_2 \to \dim \mathsf{B}_{N+1}$			
$4 \to 5$	$(1,2)$: $10 \to 2$	$(2,4)$: $9 \to 1$	$(3,1)$: $9 \to 1$	$(4,3)$: $10 \to 2$
$5 \to 6$	$(1,2)$: $9 \to 0$	$(3,1)$: $33 \to 24$	$(3,5)$: $33 \to 24$	$(5,4)$: $9 \to 0$
$6 \to 7$	$(2,3)$: $92 \to 14$	$(3,5)$: $78 \to 0$	$(4,2)$: $78 \to 0$	$(5,4)$: $92 \to 14$
$7 \to 8$	$(2,3)$: $91 \to 0$	$(4,2)$: $330 \to 239$	$(4,6)$: $330 \to 239$	$(6,5)$: $91 \to 0$

Proof. Equation (5.1) is CV's eq. (85). The operator $\psi_e^\dagger \chi_f^\dagger$ squares to zero and commutes with V_N, so the binomial expansion of $(V_N + \psi_e^\dagger \chi_f^\dagger)^n$ stops after one term. Both components are old BPS states. From (2.7) and $E|\Omega\rangle = 0$,

$$E\,V^n|\Omega\rangle = \sum_m V^m [E,V] V^{n-1-m}|\Omega\rangle = n\,K\,V^{n-1}|\Omega\rangle, \tag{5.2}$$

so $J_{10} = EV^n|\Omega\rangle - nKV^{n-1}|\Omega\rangle = 0$. For J_{01}, compute $Q_{N+1}^\dagger V_{N+1}^n|\Omega\rangle = \chi_f^\dagger J_{01}$, using $Q_0^\dagger V^m|\Omega\rangle = 0$. The left side vanishes by CCSY (3.23) at size $N+1$. The remaining six equations of Table 2 hold trivially for harmonic Ψ_{00}, Ψ_{11}. The weights follow from CV (86): $n^2\|V^{n-1}|\Omega\rangle\|^2/\|V_{N+1}^n|\Omega\rangle\|^2 = n/(N+1)$. □

[derived + verified] The bare fraction equals 1, and the slot weights agree to 4×10^{-16}, for all n at $N=2$–7.

Reading. [interpretation] The current J_{10} is what the old model must absorb when a state in slot 00 is promoted, $Q_0\Psi_{10} = J_{10}$. A fortuitous state must absorb it with a Hodge dressing cloud. The tower instead pays with a second old BPS state: the 11 component $nV^{n-1}|\Omega\rangle$, whose K-image cancels the E-image of the 00 component. The cancellation is the Ward identity of V. The tower's uplift is fixed by symmetry and involves no dynamics of the old model.

Two notions of dressing. CV call (5.1) "genuinely dressed" ($\kappa_{00} > 0$), because relative to the 00 channel the state needs its 11 component. In Hodge terms it has no dressing at all. Both statements are correct. CV's dressing measures support outside one channel; the Hodge dressing measures non-BPS content in any slot. This report says *bare* for the latter. The tower is not an eigenvector of CV's slot decoders $P\Pi_T P$: the residuals $\|P\Pi_T P\,t - \lambda t\|$ (with $\lambda = \langle t, P\Pi_T P t\rangle$ the slot weight) are 0.12–0.43 for $N=5\to6$ through $7\to8$, and equal for $T=00$ and 11 because $P\Pi_{10}Pt = P\Pi_{01}Pt = 0$.

6 Bare states: arithmetic walls and the tower excess

The bare map. With all four $\Psi_T = h_T$ harmonic, Table 2 reduces to:

- $Kh_{01} = 0$ and $Eh_{01} = 0$;
- $E^\dagger h_{10} = 0$ and $K^\dagger h_{10} = 0$;
- $J_{10} = Eh_{00} - Kh_{11} + Zh_{01} = 0$ and $J_{01} = K^\dagger h_{00} + E^\dagger h_{11} + Z^\dagger h_{10} = 0$.

The bare BPS states are the kernel of this linear map on $\bigoplus_T \mathsf{B}(\text{slot } T)$. Each equation lives in a known constrained space. Kh_{01} is closed in $(A+2, B-1)$, Eh_{01} closed in $(A+1, B-2)$, $E^\dagger h_{10}$ and $K^\dagger h_{10}$ coclosed in $(A-2, B+1)$ and $(A-3, B)$. J_{10} is closed in $(A+1, B-1)$ once $Kh_{01} = Eh_{01} = 0$, because $Q_0 J_{10} = [E, K]h_{01}$; J_{01} is coclosed in $(A-2, B)$ by the mirror argument.

Generic count. Placing each equation in generic position inside its constrained space gives $n_{01} = \max(0, b_{01} - c_1 - c_2)$ and $n_{10} = \max(0, b_{10} - c_3 - c_4)$ for the slot-01 and slot-10 kernels, where the c's are the dimensions of the constrained spaces. The count is then

$$\text{bare}_{\text{gen}} = \max\big(0,\ b_{00} + b_{11} + n_{01} + n_{10} - c(J_{10}) - c(J_{01})\big). \tag{6.1}$$

Result. [numerical] The law

$$\#\text{bare} = \text{bare}_{\text{gen}} + \big[\,1 \text{ in diagonal sectors } (n,n),\ 1 \le n \le N,\ \text{where bare}_{\text{gen}} = 0\,\big] \tag{6.2}$$

holds with no exception in all 400 symmetrized and 240 generic sector records at $N = 2$–6, and at $N = 7$ (Table 5). This uses the exact dimensions of the two sub-kernels n_{01}, n_{10}. Their generic values fail by one at $N = 5$ only (see below). The bracket is the tower. The excess first appears at $N = 6 \to 7$ in the central sectors $(3,3)$ and $(4,4)$, and at $N = 7 \to 8$ in *every* diagonal sector. The generic model never has an excess. Its bare states, the *arithmetic walls*, coincide in number with those of the symmetrized model in every sector (Fig. 2).

Table 5: Bare BPS states vs. the generic count (6.1), seed 1 (other seeds identical). Sectors not listed have neither. "tower": the excess state is the tower; the generic model has 0 there.

$N \to N+1$	sector: bare (generic count)
$6 \to 7$	$(1,0)$: 7 (7); $(1,1)$: 2 (2); $(2,0)$: 8 (8); $(2,1)$: 41 (41); $(2,2)$: 3 (3); $(3,2)$: 25 (25); $(3,3)$: **1 (0) tower**; $(4,4)$: **1 (0) tower**; and mirrors $(A,B) \to (N{+}1{-}A, N{+}1{-}B)$
$7 \to 8$	$(1,0)$: 8 (8); $(1,1)$: **1 (0) tower**; $(2,0)$: 13 (13); $(2,1)$: 56 (56); $(2,2)$: **1 (0) tower**; $(3,1)$: 7 (7); $(3,2)$: 50 (50); $(3,3)$, $(4,4)$: **1 (0) tower** each; and mirrors

The $N = 5$ coincidences. [numerical] At $N = 5$, and only there among $N = 2$–7, three generic-position counts fail by one dimension. All three failures occur in both models and in all seeds. The old-model gaps are clean (zero modes $\le 10^{-13}$, gaps ≥ 0.03), so these are not tolerance artifacts.

1. $\dim\ker[E^\dagger; K^\dagger]$ on $\mathsf{B}(2,2)$, and its mirror $\ker[K; E]$ on $\mathsf{B}(3,3)$.
2. $\ker[K; E]$ on $\mathsf{B}(2,1)$, and its mirror $\ker[E^\dagger; K^\dagger]$ on $\mathsf{B}(3,4)$.
3. $\mathsf{B}(2,2) \cap (\text{im}\, D_{01})^\perp$, and its mirror (§7).

The extra vector in item 3 is not the one in item 1 (overlap 4–6 %). We have no explanation. They recall the unexplained "atom excess" of the one-flavor model (D6 §4).

Monotone vs. fortuitous at the level of bare states. Every bare fortuitous state is an arithmetic wall. Walls are determined by counting, identical for generic couplings, and confined at $N = 7 \to 8$ to the peripheral sectors $(1,0)$, $(2,0)$, $(2,1)$, $(3,1)$, $(3,2)$ and their mirrors. The tower is the only bare state the counting does not produce.

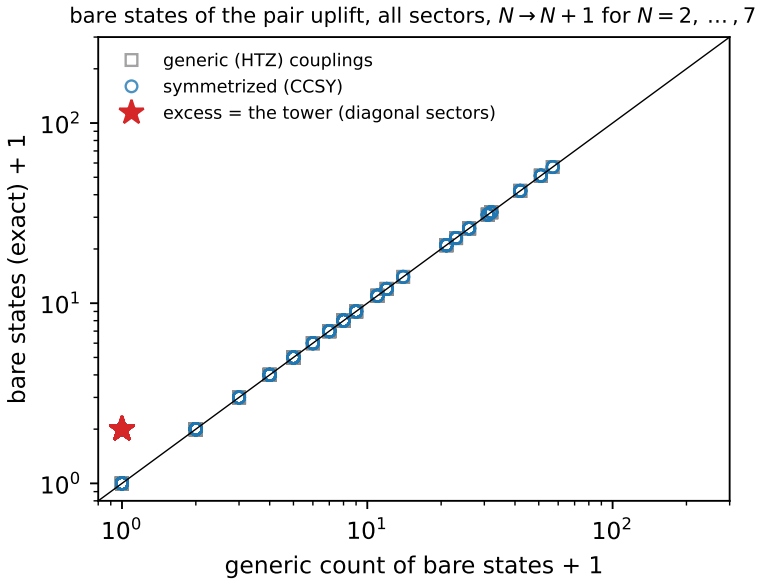


Figure 2: Bare BPS states of the pair uplift against the generic count (6.1), for all sectors and realizations at $N = 2 \to 3, \ldots, 7 \to 8$. Symmetrized (circles) and generic (squares) couplings coincide everywhere. The only points off the diagonal are the tower (stars): a bare state in a diagonal sector where the count predicts none. [numerical]

7 CV's channel decoders: rooms and generic position

CV's decoder for channel T maps an enlarged BPS state to its slot-T component, $D_T = \Pi_T|_{\mathsf{B}_{N+1}}$. An old BPS state α lifts exactly through T if $\alpha \in \operatorname{im} D_T$. The *exact-lift locus* is $A_T = \mathsf{B} \cap \operatorname{im} D_T$, and U_T is the minimum fidelity $\|P_{\operatorname{im} D_T}\alpha\|^2$ over the unit fortuitous states (CV eqs. 44, 50, 89).

7.1 Constrained rooms

CV §7.1 explained the one-flavor exact-lift loci by placing B and $\operatorname{im} D_T$ in generic position inside a constrained "room", the kernel of Q_N or $Q_N^\dagger$. Table 2 gives the analogous rooms for the four channels. [derived] For the old sector (a, b):

$$\begin{aligned} R_{01} &= \ker Q_0, \qquad R_{10} = \ker Q_0^\dagger, \\ R_{00} &= \mathsf{B}(a,b) \oplus Q_0^+ K\big(\ker Q_0|_{(a,b-1)}\big) \oplus (Q_0^\dagger)^+ E^\dagger\big(\ker Q_0^\dagger|_{(a-1,b)}\big), \\ R_{11} &= \mathsf{B}(a,b) \oplus Q_0^+ E\big(\ker Q_0|_{(a+1,b)}\big) \oplus (Q_0^\dagger)^+ K^\dagger\big(\ker Q_0^\dagger|_{(a,b+1)}\big). \end{aligned} \tag{7.1}$$

For 01 and 10 this is the closedness of Ψ_{01} and the coclosedness of Ψ_{10}. For 00 it follows from the first row of Table 2. The coexact part of Ψ_{00} is $-Q_0^+ K \Psi_{01}$ with Ψ_{01} closed, and its exact part is $-(Q_0^\dagger)^+ E^\dagger \Psi_{10}$ with Ψ_{10} coclosed. Channel 11 follows likewise.

7.2 Two counting laws

Law 1 (rank). *[numerical]* $\operatorname{rk} D_T = \min(d_{\text{new}}, R_T)$, *with* $d_{\text{new}} = \dim \mathsf{B}_{N+1}$ *of the target sector.*

Law 2 (exact-lift locus). *[numerical]* $\dim A_T = \max(0,\ b + \operatorname{rk} D_T - R_T)$.

Both laws hold with no exception in 1300 (sector, channel, seed) records of the symmetrized model and 780 of the generic model at $N = 2$–6, and in the further 176 and 88 records at $N = 7$. They imply three corollaries for the fortuitous subspace F, of dimension f:

(C1) $U_T = 1$ if and only if $d_{\text{new}} \geq R_T$. On the diagonal this uses the fact that the tower lies in A_{00} and A_{11}.

(C2) $U_T = 0$ if and only if $F \cap (\operatorname{im} D_T)^\perp \neq 0$. The generic count is $b - \operatorname{rk} D_T > 0$ for the channels 00 and 11 on the diagonal, where the tower is protected, and $f - \operatorname{rk} D_T > 0$ otherwise.

(C3) $0 < U_T < 1$ in all remaining cases.

7.3 CV's Fig. 6, reproduced and explained

Figure 3 shows U_T for the fortuitous subspaces in CV's eleven sectors at $N = 2 \to 3, \ldots, 7 \to 8$, computed with our code, together with the category predicted by (C1)–(C3). [numerical] The observations are:

- The 0/1 pattern is CV's. The intermediate values agree within realization scatter, e.g. $(1,2)/00$ at $3 \to 4$: 0.30 here, 0.45 in CV.
- The predicted category agrees with the observation in every cell and every seed of the symmetrized model.
- In the generic control, the categories fail only at $N = 5$ in $(2,2)/01$ and its mirror (coincidence 3 above).

Some intermediate values coincide, e.g. $(1,2)/00$ and $(2,1)/11$ at $3 \to 4$, in CV's table as well. These are particle–hole symmetry, $(a, b; T) \leftrightarrow (N - a, N - b; \bar{T})$, acting realization by realization.

CV's qualitative summary reads "the charge-preserving channel 00 fails for the χ-heavy sectors, whereas those sectors lift perfectly through the $+\psi$ channel". In our terms it is the inequality $d_{\text{new}} \geq R_T$, sector by sector. Table 6 shows the numbers for the χ-heavy rows. For χ-heavy sectors, $+\psi$ moves the target toward the diagonal, where BPS states are plentiful, so d_{new} exceeds the room. The 00 channel keeps the charge, and its target BPS space falls below the room as N grows.

7.4 The tower in CV's channels, and its dressing cost

The tower lies in A_{00} and A_{11} for every N: a specific vector of B lies in a subspace of positive codimension, so it is not in generic position. Yet $\dim A_T$ still equals the generic count; the tower is one of its directions. In the single-mode channels the tower behaves like any vector of B. Its fidelities there are reproduced together with CV's Fig. 7 (their eq. 90 is again particle–hole symmetry).

CV's dressing cost is $\kappa_T(\alpha) = \|D_T^+ \alpha\|^2 / \|\alpha\|^2 - 1$, with $D_T^+ \alpha$ the minimum-norm exact uplift. For the tower, (85) supplies one uplift with cost $n/(N+1-n)$ in channel 00 (CV 87) and $(N-n)/(n+1)$ in channel 11 (CV 88). The minimum-norm uplift is the tower minus its projection onto $\ker D_T$, so the cost reaches the bound if and only if the tower is orthogonal to $\ker D_T$. [numerical] In all 255 (rung, channel, seed) records at $N = 2$–7, the bound is attained *exactly when* $\ker D_T = 0$, and the cost lies strictly below it otherwise (Fig. 4). Combined with Law 1, $\ker D_T = 0 \iff d_{\text{new}} \leq R_T$, so

$$\begin{aligned} \kappa_{00}(V^n|\Omega\rangle) = \frac{n}{N+1-n} \quad &\iff \quad d_{\text{new}}(n,n) \leq R_{00}(n,n), \\ U_{00}(n,n) = 1 \quad &\iff \quad d_{\text{new}}(n,n) \geq R_{00}(n,n). \end{aligned} \tag{7.2}$$

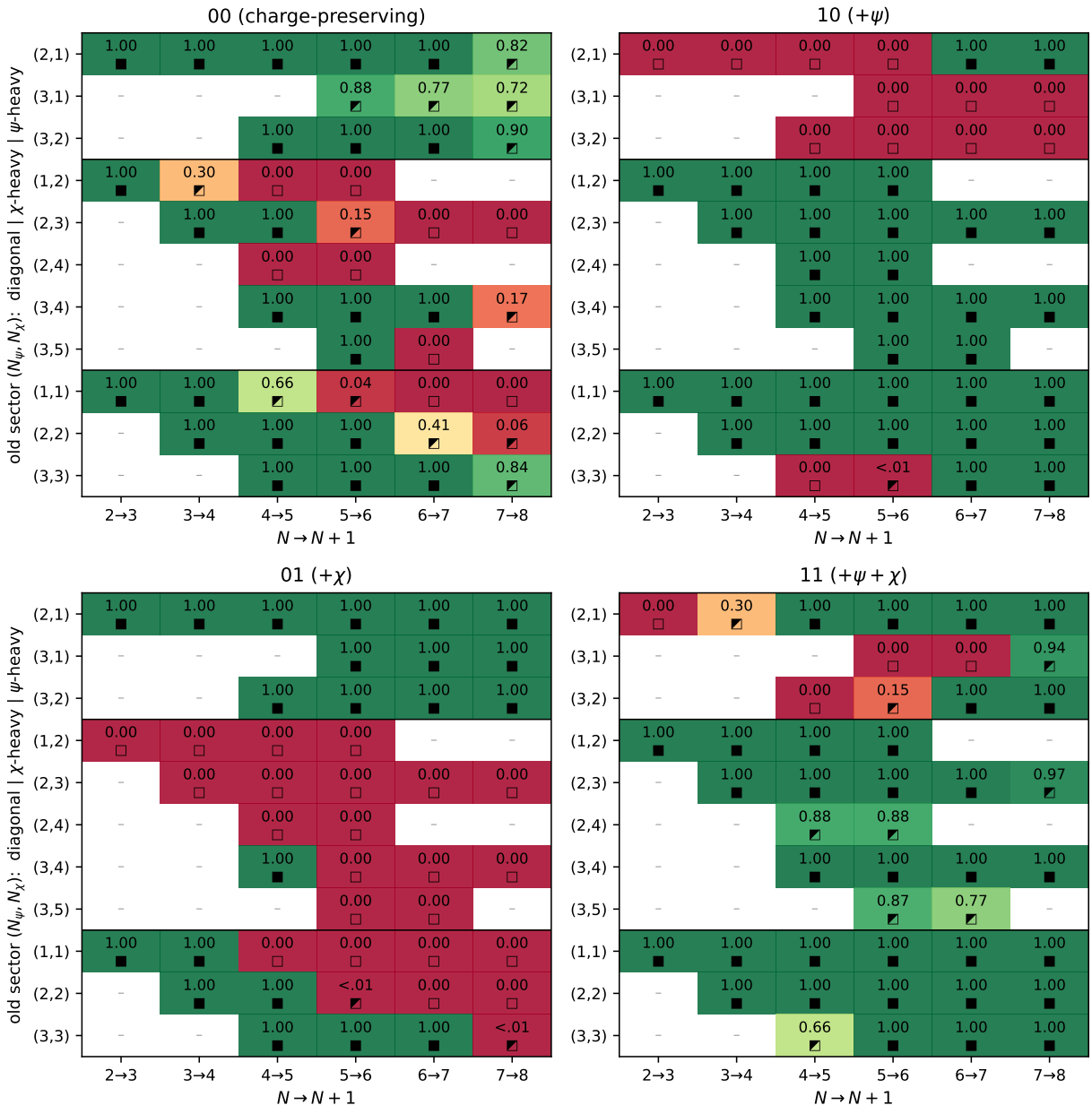


Figure 3: CV's Fig. 6 recomputed: U_T of the fortuitous subspace by sector (rows) and channel (panels), minimum over seeds (five for $N \le 6$, two for $N = 7$). Symbols give the category predicted by Laws 1–2: ■ $U = 1$ (rk $D_T = R_T$), half-filled $0 < U < 1$, □ $U = 0$. No cell disagrees. "< .01" marks small positive values.

Table 6: The mechanism behind CV's χ-heavy rows (seed 1, $N = 3$–7): target BPS dimension d_{new} vs. room R_T, the exact-lift codimension $b - \dim A_T$, and U_T. $U_T = 1$ exactly when $d_{\text{new}} \geq R_T$.

		channel 00				channel 10 ($+\psi$)			
N	sector	d_{new}	R_{00}	codim	U	d_{new}	R_{10}	codim	U
3	$(1,2)$	8	9	1	0.39	28	9	0	1
3	$(2,3)$	17	3	0	1	12	3	0	1
4	$(1,2)$	2	20	8	0	65	24	0	1
4	$(2,3)$	45	24	0	1	65	23	0	1
4	$(2,4)$	1	6	2	0	35	6	0	1
4	$(3,4)$	35	4	0	1	15	4	0	1
5	$(1,2)$	0	17	2	0	116	50	0	1
5	$(2,3)$	66	100	34	0.30	220	95	0	1
5	$(2,4)$	0	47	1	0	144	49	0	1
5	$(3,4)$	144	50	0	1	116	45	0	1
5	$(3,5)$	24	10	0	1	60	10	0	1
6	$(2,3)$	14	217	66	0	546	285	0	1
6	$(3,4)$	364	300	0	1	546	264	0	1
6	$(3,5)$	0	120	24	0	344	114	0	1
7	$(2,3)$	0	252	14	0	1072	700	0	1
7	$(3,4)$	560	1205	364	0.21	1820	1078	0	1

The tower's cost locks onto CV's bound exactly when the charge-preserving channel stops lifting every fortuitous state of the sector; both happen at $d_{\text{new}} = R_{00}$ when equality holds. This explains CV's observation that the cost "rises onto [the bound] only once $N \gtrsim n + 3$". Here $n = 1$ is at the bound from $N = 4$, $n = 2$ from $N = 6$, and $n = 3$ from $N = 7$, while $n = 4$ is still below it at $N = 7$. Sector $(3,3)$ shows both halves of (7.2) across two sizes. At $N = 6 \to 7$ the tower's cost is below the bound and $U_{00} = 1$. At $N = 7 \to 8$ the cost sits on the bound and $U_{00} = 0.84$ (Fig. 3).

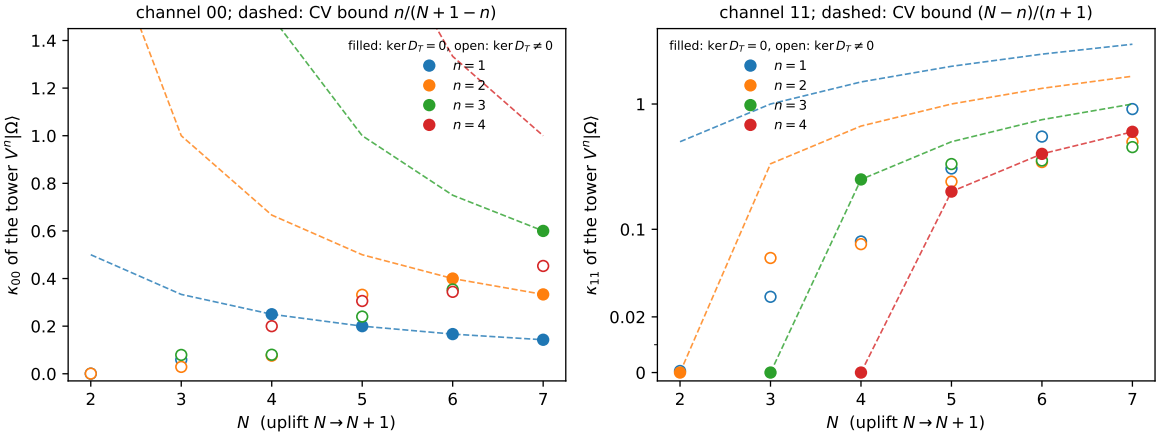


Figure 4: CV dressing costs of the tower, computed as exact minimum-norm uplifts (seed 1), against CV's bounds (dashed). Filled markers: $\ker D_T = 0$, where the cost equals the bound. Open: $\ker D_T \neq 0$, where it lies below. [numerical]

8 First order: dressing as interfering two-point functions

First-order dressing. Replace each Ψ_T by its harmonic source h_T on the right of Table 2. The six non-harmonic pieces of a BPS state are then

$$\begin{aligned} z_{10} &= Q_0^+ J_{10}(h), & x_{01} &= (Q_0^\dagger)^+ J_{01}(h), & z_{00} &= -Q_0^+ K h_{01}, \\ x_{00} &= -(Q_0^\dagger)^+ E^\dagger h_{10}, & z_{11} &= -Q_0^+ E h_{01}, & x_{11} &= (Q_0^\dagger)^+ K^\dagger h_{10}. \end{aligned} \tag{8.1}$$

The first-order dressing cost $\kappa^{(1)}$ is the sum of their squared norms divided by $\|h\|^2$. The first-order bare fraction is $1/(1+\kappa^{(1)})$, obtained as a generalized eigenproblem on the admissible sources. As in one flavor, an exactly bare state has $\kappa^{(1)} = 0$, and numerically the converse holds as well.

Single-slot sources: two-point functions. [derived + verified] On exact forms Q_0^+ has norm-square $\langle j, Gj\rangle$, and likewise $(Q_0^\dagger)^+$ on coexact forms, with $G = H_0^+ = \int_0^\infty d\tau\, e^{-\tau H_0}$. Hence a source in one slot has

$$\begin{aligned} \sigma_{00}^2 &= \langle Eh, GEh\rangle + \langle K^\dagger h, GK^\dagger h\rangle, & \sigma_{01}^2 &= \langle Kh, GKh\rangle + \langle Eh, GEh\rangle + \langle Zh, GZh\rangle, \\ \sigma_{11}^2 &= \langle Kh, GKh\rangle + \langle E^\dagger h, GE^\dagger h\rangle, & \sigma_{10}^2 &= \langle E^\dagger h, GE^\dagger h\rangle + \langle K^\dagger h, GK^\dagger h\rangle \\ & & &\quad + \langle Z^\dagger h, GZ^\dagger h\rangle. \end{aligned} \tag{8.2}$$

These hold to 10^{-8} (test). Each term is a zero-temperature, Euclidean-time-integrated two-point function in the old model, between a BPS state and the continuum, of one of three matter operators: flavor conversion $\psi^\dagger\chi$ (E), ψ-pair creation (K), or single-ψ creation (Z). This generalizes D6's $\sigma^2 \in \operatorname{spec} PE^\dagger H_0^{-1} EP$. For terms involving Z, J_{10} need not be closed, and the pseudo-inverse projects first.

Interference. Sources in several slots share the currents (2.8). The dressing therefore contains cross terms between different matter operators, e.g. $-2\,\mathrm{Re}\langle Eh_{00}, G\,Kh_{11}\rangle$ from $\|Q_0^+ J_{10}\|^2$. [derived + verified] The tower is the exactly dark state of this quadratic form: $J_{10} = J_{01} = 0$, although its incoherent single-slot sum is not small. That sum is 0.07–6.9 at $N = 5$ and 0.07–1.8 at $N = 6$. It varies between realizations with the old model's gap, which is as small as 0.03 at $N = 5$.

[numerical] Fortuitous states interfere too (Fig. 5b). In the eigenbasis of the exact bare fraction, the ratio $(\kappa^{(1)} - \kappa_{\mathrm{inc}})/\kappa_{\mathrm{inc}}$ decreases with μ. It is near $+1$ (constructive) for the most dressed states and approaches -1 (complete cancellation) as $\mu \to 1$. At $N = 6 \to 7$ it is close to a single function of μ. At $N = 7 \to 8$ a second, more destructive branch appears at small μ. Part of this correlation is variational: the decoder's eigenvectors extremize the dressing form, so extreme μ goes with extreme interference. It nevertheless shows that the spread of the bare-fraction spectrum comes largely from interference between slots, not from the spread of single-slot costs. **We therefore retract the working hypothesis "monotone = coherent, fortuitous = incoherent".** What distinguishes the tower is *exact* cancellation at every N, enforced by an identity, not the presence of interference.

Accuracy. [numerical] First order reproduces the exact bare-fraction distribution (Fig. 5a). Sector means agree to within 4.2 % in all sectors with at least 50 states at $N = 5, 6$. First order is usually lower, e.g. 0.7014 vs. 0.7322 in $(3,3)$ at $N = 6 \to 7$, five seeds. The agreement degrades slowly with N: at $N = 7 \to 8$ (seed 1) it is 0.6713 vs. 0.7150 in the central sector $(4,4)$, and all sectors lie within 6.2 %. Walls coincide in every sector. Exact walls are necessarily first-order walls; the converse is observed, not proved.

Near-bare fortuitous states and the hard edge. [numerical] States with μ close to 1 are numerous at $N = 6 \to 7$. The fraction with $0.99 < \mu < 1$ is 0.16 in the central sectors. There the bare map is nearly square: 336 unknowns against 339 constraints in $(3, 3)$, so bare_{gen} sits three short of threshold. This is the analogue of D6's hard edge. At $N = 7 \to 8$ the bare map is far from square (1092 unknowns against 1428 constraints in $(4, 4)$), and the near-bare fraction falls to 0.065 (seed 1). Going down in N it is 0.07–0.08 at $N = 5$ and 0.003 at $N = 4$, where unknowns exceed constraints and exact walls take over. We had seen partial $N = 7$ quantiles before writing this, so it is a consistency check, not a blind test.

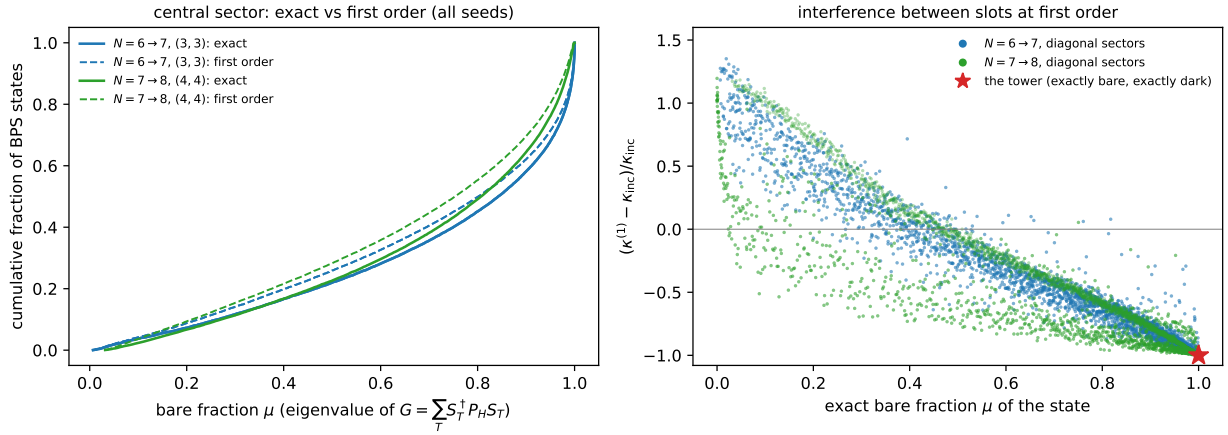


Figure 5: (a) Cumulative distribution of the bare fraction μ in the central sector, exact (solid) vs. first order (dashed). (b) Interference between slots at first order, $(\kappa^{(1)} - \kappa_{\text{inc}})/\kappa_{\text{inc}}$, against the exact bare fraction for every dressed state of the diagonal sectors. The tower (star) is exactly bare and exactly dark. [numerical]

9 The exact uplift in closed form

Added 25 October. Sections 3–4 solved the pair uplift through an intermediate model M, the old model plus ψ_e alone, whose Hodge theory is not old-model data. This section removes M. Every object below is built from the old model's BPS bases, its pseudo-inverses Q_0^+, $(Q_0^\dagger)^+$, its Green's operator $G = H_0^+$, and the three couplings E, K, Z.

Notation.

- Slot vectors are ordered by w: $\Psi = (\Psi_{01}, \Psi_{00}, \Psi_{11}, \Psi_{10})$.
- The new supercharge splits as $Q_{N+1} = Q_d + Q_c$. The slot-diagonal part is $Q_d = \text{diag}(-Q_0, Q_0, Q_0, -Q_0)$. The cross-slot part Q_c carries the arrows of Fig. 1.
- P_H is the slotwise old harmonic projector, and $E_1 = \bigoplus_T \mathsf{B}(\text{slot } T)$ is the old BPS data.

9.1 Cohomology from old data: homological perturbation

The triple (inclusion of E_1, projection P_H, homotopy $h = -Q_d^+$) is a strong deformation retract of the slotwise complex. That is, $Q_d h + h Q_d = \iota P_H - 1$ and $h^2 = h\iota = P_H h = 0$. The cross-slot part Q_c raises w while h preserves it, so $(hQ_c)^3 = 0$. The homological perturbation lemma therefore applies with *finite* series.

Theorem 2 (Cohomology from old data)**.** *[derived + verified] $H(Q_{N+1}) \cong H(E_1, D)$, with the transferred differential $D = P_H Q_c (1 + Q_d^+ Q_c)^{-1} P_H$:*

$$D\eta = \big(0,\ P_H K \eta_{01},\ P_H E \eta_{01},\ P_H(E\eta_{00} - K\eta_{11} + \mathcal{Z}\eta_{01})\big), \qquad \mathcal{Z} = Z - E\,Q_0^+ K + K\,Q_0^+ E. \quad (9.1)$$

Hence $\dim \mathsf{B}_{N+1}(A,B) = \dim E_1(A,B) - \operatorname{rk} D_{(A,B)} - \operatorname{rk} D_{(A-2,B+1)}$.

Proof. The perturbation lemma gives $D = P_H Q_c (1 - h Q_c)^{-1} \iota$, and the series stops at second order. The $w{+}1$ part of D is the slot complex d_1 of (4.4). The $w{+}2$ part is $d_2 = P_H(Z - EQ_0^+ K + KQ_0^+ E)P_H = P_H \mathcal{Z} P_H$, and there are no higher terms. As a consistency check, the only component of D^2 is $P_H(EP_H K - KP_H E)P_H = P_H[E,K]P_H = P_H\{Q_0, Z\}P_H = 0$. The first equality uses the fact that $(1 - P_H)K$ maps harmonic forms to exact ones and E keeps them exact. □

[derived + verified] There are no mismatches in any sector record: 950 symmetrized and 570 generic at $N \le 6$, and 81 of each model at $N = 7$. The d_2 of Table 4 is therefore no longer an obstacle: the old BPS data together with $\mathcal{Z}$ determine the enlarged BPS space exactly. $\mathcal{Z}$ is a secondary, Massey-type operation. A bottom-slot class that d_1 annihilates reaches the top slot either directly through the single-ψ coupling Z, or by zig-zags through the dressed middle slots, $EQ_0^+ K$ and $KQ_0^+ E$.

9.2 BPS states in closed form

Theorem 3 (Closed-form BPS states)**.** *[derived + verified] Let $\mathcal{K} = Q_d^+ Q_c + (Q_d^\dagger)^+ Q_c^\dagger$. A slot vector Ψ is BPS if and only if*

$$(1 + \mathcal{K})\Psi = P_H \Psi \qquad \text{and} \qquad P_H Q_c \Psi = 0 = P_H Q_c^\dagger \Psi. \quad (9.2)$$

If $1 + \mathcal{K}$ is invertible, then $\mathsf{B}_{N+1} = \{(1+\mathcal{K})^{-1} h :\ h \in \mathcal{A}\}$ with $\mathcal{A} = \ker P_H (Q_c \oplus Q_c^\dagger)(1+\mathcal{K})^{-1}|_{E_1}$, and the four-slot harmonic map is injective.

Proof. ($\Rightarrow$) Write $\Psi = h + r$ slotwise. $Q_{N+1}\Psi = 0$ gives $Q_d r = -Q_c \Psi$. So $Q_c \Psi$ is exact, and the coexact part of r is $-Q_d^+ Q_c \Psi$. Dually, the exact part of r is $-(Q_d^\dagger)^+ Q_c^\dagger \Psi$. Together these say $r = -\mathcal{K}\Psi$.

($\Leftarrow$) From $(1+\mathcal{K})\Psi = P_H \Psi$, $Q_{N+1}\Psi = (P_H + P_{\mathrm{co}})u$ with $u = Q_c \Psi$. Nilpotency gives $Q_d u = -Q_c(P_H + P_{\mathrm{co}})u$. Since Q_c raises w, the lowest-w component of u is Q_d-closed. Its coexact part therefore vanishes, and its harmonic part vanishes by assumption, so it is exact. Proceeding upward in w shows $Q_{N+1}\Psi = 0$. The adjoint condition follows by downward induction. □

The ten blocks of $\mathcal{K}$ are explicit, e.g. $(00 \leftarrow 01) = Q_0^+ K$, $(10 \leftarrow 00) = -Q_0^+ E$, $(01 \leftarrow 10) = -(Q_0^\dagger)^+ Z^\dagger$. [derived + verified] At $N \le 6$, in both models and all seeds:

- $(1+\mathcal{K})^{-1}$ applied to the harmonic slot parts of each BPS state reconstructs it to 2×10^{-13};
- $\dim \mathcal{A} = \dim \mathsf{B}_{N+1}$ in every sector;
- the closed-form states are BPS to 10^{-13}.

9.3 Integrating out the middle slots

The dressing of the middle slots is explicit in the outer components: $z_{00} = -Q_0^+ K \Psi_{01}$, $x_{00} = -(Q_0^\dagger)^+ E^\dagger \Psi_{10}$, $z_{11} = -Q_0^+ E \Psi_{01}$ and $x_{11} = (Q_0^\dagger)^+ K^\dagger \Psi_{10}$. Substituting these leaves two equations of Dyson type for the outer slots.

Proposition 3 (Reduced system). *[derived + verified] The outer components of a BPS state satisfy*

$$P_{\mathrm{Co}}(1+\mathcal{S}_G)\Psi_{10} = Q_0^+\big(Eh_{00}-Kh_{11}+\mathcal{Z}\Psi_{01}\big), \qquad P_{\mathrm{Ex}}(1+\mathcal{S}'_G)\Psi_{01} = (Q_0^\dagger)^+\big(K^\dagger h_{00}+E^\dagger h_{11}+\mathcal{Z}'^\dagger\Psi_{10}\big), \tag{9.3}$$

with the self-energies

$$\mathcal{S}_G = E\,G\,E^\dagger + K\,G\,K^\dagger \quad (\textit{slot } 10), \qquad \mathcal{S}'_G = K^\dagger G\,K + E^\dagger G\,E \quad (\textit{slot } 01). \tag{9.4}$$

$\mathcal{Z}'$ denotes $\mathcal{Z}$ one charge level down. For the unknown dressing $u = (z_{10}, x_{01})$ this is $\mathbb{M}u = \sigma(h)$, with

$$\mathbb{M} = \begin{pmatrix} 1+\mathsf{A} & -L \\ -L' & 1+\mathsf{A}' \end{pmatrix}, \qquad \mathsf{A} = P_{\mathrm{Co}}\mathcal{S}_G P_{\mathrm{Co}} \geq 0, \quad \mathsf{A}' = P_{\mathrm{Ex}}\mathcal{S}'_G P_{\mathrm{Ex}} \geq 0, \quad L = Q_0^+\mathcal{Z}, \quad L' = (Q_0^\dagger)^+\mathcal{Z}'^\dagger. \tag{9.5}$$

The two off-diagonal blocks are related by $L' = -L^\dagger + C$, where the leak

$$C = \alpha_{00}\gamma_{00}^\dagger - \alpha_{11}\gamma_{11}^\dagger, \qquad \alpha_{00} = (Q_0^\dagger)^+K^\dagger\big|_{\mathsf{B}_{00}}, \quad \gamma_{00} = Q_0^+E\big|_{\mathsf{B}_{00}}, \quad \alpha_{11} = (Q_0^\dagger)^+E^\dagger\big|_{\mathsf{B}_{11}}, \quad \gamma_{11} = Q_0^+K\big|_{\mathsf{B}_{11}}, \tag{9.6}$$

factors through the BPS spaces B_{00}, B_{11} of the middle slots.

Sketch. The self-energy form of A follows from $\langle z, Q_0^+E(Q_0^\dagger)^+E^\dagger z\rangle = \|P_{\mathrm{ex}}E^\dagger v\|^2 = \langle z, EGE^\dagger z\rangle$ for $z = Q_0^\dagger v$, and the same with $E \to K$. Both forms are Hermitian, so they agree. For the leak, $\{Q_0, Z\} = [E, K]$ together with $Q_0Q_0^+ = P_{\mathrm{ex}}$ and $Q_0^+Q_0 = P_{\mathrm{co}}$ gives $\{Q_0, \mathcal{Z}\} = EP_HK - KP_HE$. For $x = Q_0w$ and $z = Q_0^\dagger v$ this yields $\langle z, Lx\rangle + \langle L'z, x\rangle = \langle v, (EP_HK - KP_HE)w\rangle = \langle z, (\gamma_{00}\alpha_{00}^\dagger - \gamma_{11}\alpha_{11}^\dagger)x\rangle$. □

[interpretation] The middle slots act as virtual channels. Integrating them out leaves the outer slots with self-energies, which are Euclidean-time-integrated two-point functions of E and K through the old continuum, and a transfer amplitude $\mathcal{Z}$. $\gamma_{00}h$ and $\alpha_{00}h$ are the first-order top- and bottom-slot dressings that a middle-slot BPS state h would create. The leak is the path "top dressing → middle BPS state → bottom dressing". It is absent when the middle slots carry no BPS states. [derived + verified] A, A' equal the projected self-energies, and $L' + L^\dagger = C$ holds with the factorized C, both to 10^{-13}.

9.4 When is the uplift unique?

Proposition 4 (Uniqueness criterion). *[derived]* $\operatorname{Herm}\mathbb{M} = 1+\mathsf{A}\oplus\mathsf{A}' - \frac{1}{2}\begin{pmatrix} 0 & C^\dagger \\ C & 0 \end{pmatrix}$, *so* $\operatorname{Herm}\mathbb{M} > 0$ *if and only if $\rho := \|(1+\mathsf{A}')^{-1/2}C(1+\mathsf{A})^{-1/2}\| < 2$. Then $\mathbb{M}$ is accretive and hence invertible. So are $1+\mathcal{K}$ and the four-slot harmonic map, because the middle rows of $1+\mathcal{K}$ are explicit. Without middle cohomology, $C = 0$, $\operatorname{Herm}\mathbb{M} \geq 1$ and $\|\mathbb{M}^{-1}\| \leq 1$, which is the contraction of the one-flavor uplift.*

[numerical] At unit coupling the criterion holds in 471 of 499 reduced systems (symmetrized) and 299 of 319 (generic) at $N \leq 7$. $\mathbb{M}$ was invertible in all of them. At $N \leq 6$ the smallest $\sigma_{\min}(\mathbb{M})$ is 0.31 and 0.11. At $N = 7$ (seed 1) it falls to 0.013 in sectors $(3,3)$ and $(5,5)$ of the symmetrized model, and to 0.051 in the generic one. The old model there has a near-zero mode (gap 0.0032 in sector $(2,3)$), which inflates the dressing to $\|\mathcal{K}\| \approx 150$. This is the origin of the bare fraction 1.8×10^{-6} noted in §4.

Injectivity is generic, not universal. $\det \mathbb{M}$ is complex, so it can vanish on a set of codimension two in coupling space. We scaled two independent families of the couplings that involve the new pair and minimized $\sigma_{\min}(\mathbb{M})$. This finds exact singular points in both models (Table 7). At these points the BPS dimension is unchanged, yet one BPS state has bare fraction zero: all four of its slot components are non-BPS states of the old model. We call these *pure-dressing* BPS states. In the symmetrized example the tower of the same sector remains exactly bare. Four-slot injectivity, listed as an open problem in the first version of this report, is therefore false in general. The correct statements are the criterion above and the two-step parametrization of §4, which is a bijection everywhere.

Table 7: Exact pure-dressing BPS states at rescaled couplings, $N = 4 \to 5$. Generic: g_1 scales C_{Nik}, C_{iNk} (the E couplings) and g_2 scales C_{ijN} (the K couplings). Symmetrized: g_1 scales $C_{N\cdot\cdot}$ and g_2 scales $C_{\cdot N\cdot} = C_{\cdot\cdot N}$.

model	seed	sector	(g_1, g_2)	$\sigma_{\min}(\mathbb{M})$	$\dim \mathsf{B}_{N+1}$	min. bare fraction	deficit
generic	2	(3, 3)	(0.4216, 12.58)	1.8×10^{-16}	65	≈ 0	1
symm.	1	(2, 2)	(7.280, 12.30)	1.6×10^{-16}	65	≈ 0	1

"Deficit" is the injectivity deficit of the four-slot harmonic map. In the symmetrized example the tower of the same sector keeps bare fraction 1. Found and reproduced by `scripts/find_pure_dressing_states.py`.

9.5 Admissibility, slot weights and decoders

Proposition 5 (Admissibility and decoders)**.** *[derived + verified] For $h \in E_1$, $\Psi = (1+\mathcal{K})^{-1}h$ is BPS if and only if three conditions hold:*

(a) $P_H K h_{01} = P_H E h_{01} = 0$ *and* $P_H E^\dagger h_{10} = P_H K^\dagger h_{10} = 0$*;*

(b) $P_H(E h_{00}^+ - K h_{11}^+ + \mathcal{Z} h_{01}) = 0$*, with* $h_{00}^+ = h_{00} + \alpha_{00}^\dagger x_{01}$ *and* $h_{11}^+ = h_{11} + \alpha_{11}^\dagger x_{01}$*;*

(c) $P_H(K^\dagger h_{00}^- + E^\dagger h_{11}^- + \mathcal{Z}'^\dagger h_{10}) = 0$*, with* $h_{00}^- = h_{00} + \gamma_{00}^\dagger z_{10}$ *and* $h_{11}^- = h_{11} - \gamma_{11}^\dagger z_{10}$*.*

Condition (b) says that the perturbation-lemma class representative $(h_{01}, h_{00}^+, h_{11}^+, \cdot)$ of the state is a D-cocycle; (c) is the dual statement. The slot weights of a BPS state are

$$\begin{aligned} \|\Psi_{01}\|^2 &= \|h_{01}\|^2 + \|x_{01}\|^2, \qquad \|\Psi_{10}\|^2 = \|h_{10}\|^2 + \|z_{10}\|^2, \\ \|\Psi_{00}\|^2 &= \|h_{00}\|^2 + \langle \Psi_{01}, K^\dagger G K \Psi_{01}\rangle + \langle \Psi_{10}, E G E^\dagger \Psi_{10}\rangle, \\ \|\Psi_{11}\|^2 &= \|h_{11}\|^2 + \langle \Psi_{01}, E^\dagger G E \Psi_{01}\rangle + \langle \Psi_{10}, K G K^\dagger \Psi_{10}\rangle, \end{aligned} \tag{9.7}$$

and its dressing cost is $\kappa = \big[\|u\|^2 + \langle \Psi_{10}, \mathcal{S}_G \Psi_{10}\rangle + \langle \Psi_{01}, \mathcal{S}'_G \Psi_{01}\rangle\big]/\|h\|^2$. The bare-fraction decoder and CV's slot decoders are the generalized eigenvalue problems of $(\|h\|^2, \|\Psi\|^2)$ and $(\|\Psi_T\|^2, \|\Psi\|^2)$ on $\mathcal{A}$.

The same four maps α, γ govern the leak (9.6) and the corrections in (b), (c). The first-order theory of §8 is the leading term: it drops the $\mathbb{M}$ corrections and replaces Ψ_{10}, Ψ_{01} by h_{10}, h_{01} inside κ. Two special cases are worth recording.

- One added mode: there are no middle slots, so $\mathsf{A} = \mathsf{A}' = 0$ and $C = 0$. $\mathbb{M}$ reduces to $1+$ (anti-Hermitian), the one-flavor system.
- The tower: $\mathcal{K}\eta = 0$ and $\sigma(\eta) = 0$ for $\eta = (0, V^n|\Omega\rangle, nV^{n-1}|\Omega\rangle, 0)$, so $\Psi = \eta$ at every coupling.

[derived + verified] Conditions (a)–(c) give the same subspace as Theorem 3 in every sector at $N \leq 6$. The pencils reproduce the exact bare-fraction and slot-decoder spectra to 7×10^{-13}. At $N = 7$ (one seed per model) all of these checks hold to 2×10^{-11}; the larger residuals come from the nearly singular sectors of §9.4.

Most of the exact correction is self-energy. [numerical] The closed form separates three levels of approximation. *First order* uses first-order sources and sets $\mathbb{M} \to 1$ (§8). *Self-dressed* keeps the exact sources and the self-energies, $\mathbb{M} \to 1 + \mathsf{A} \oplus \mathsf{A}'$, but drops the transfer and the leak. *Exact* solves the full system. On the exact admissible sources (sectors with at least 50 BPS states, $N = 4$–6), self-dressing moves the mean bare fraction from its first-order value 56–92 % of the way to the exact value. The median is 0.67 (symmetrized) and 0.79 (generic). For example, in the central sector $(3, 3)$ at $N = 6 \to 7$ (five seeds) the three levels are 0.7014, 0.7220 and 0.7322. The direction is not always upward: in $(4, 3)$ at $N = 5 \to 6$ first order is slightly high, and self-dressing corrects that too. The self-energies are pure time-integrated two-point functions of E and K, so a chord computation of those alone would capture most of the exact correction. (`run_two_flavor_uplift.py --tasks levels`.)

10 Monotone versus fortuitous

Table 8 collects what the exact uplift says about the two kinds of states.

Table 8: What the uplift distinguishes.

	tower (monotone)	fortuitous
Hodge dressing	zero at every N (bare), [derived + verified]	positive, except arithmetic walls in peripheral sectors, [numerical]
mechanism	$J_{10} = J_{01} = 0$ by $[E, V] = K$	partial interference of E, K, Z two-point functions
slot content	00 and 11 only; weights $\frac{N+1-n}{N+1}$, $\frac{n}{N+1}$	all admissible slots
bare-state count	the only excess over the generic count	equal to the generic count, same as generic couplings
CV channels 00, 11	always exact (non-generic position)	exact iff $d_{\text{new}} \geq R_T$ (generic position)
CV cost κ_{00}	$= n/(N + 1 - n)$ iff $d_{\text{new}} \leq R_{00}$	broad, growing (CV Fig. 8)
first-order dressing form	exactly dark	$O(1)$; interference from $+1$ to ≈ -1
bare fraction over couplings	$= 1$ for all couplings	in $[0, 1)$; $= 0$ (pure dressing) on a codimension-two set

[interpretation] The two kinds of states differ in *why* they survive the uplift. The tower survives by symmetry. Its uplift is kinematical: old BPS states rearranged between slots, with all dressing currents cancelling by a Ward identity of the $\mathfrak{su}(2)$ that generates it. CCSY call the V's the analogue of gravitons, and the tower is the analogue of a multi-graviton state built on the vacuum. Fortuitous states survive or fail by arithmetic, and when they survive they are dressed. Their dressing is set by the old model's dynamics through Euclidean-time-integrated two-point functions. The symmetrized and generic models are indistinguishable in all counting statistics, so the fortuitous sector does not "know" about the symmetry. Only the one state per diagonal sector that the symmetry singles out behaves differently.

11 Discussion

11.1 What we have learned

The one-flavor uplift at half filling was unobstructed: every old BPS state had an uplift, with a computable dressing cloud. The two-flavor uplift is obstructed. Theorem 1 shows that, once the obstruction is imposed on the harmonic sources, the rest of the D6 structure survives intact: the anti-Hermitian system, the orthogonal splitting and the minimum-norm uplift. The pair is two such steps. It also carries a genuinely new feature, the d_2 differential, so the enlarged BPS space is not a function of old cohomology and first-order maps alone. The closed forms of §9 show what it *is* a function of: the old BPS data; the self-energies $EGE^\dagger + KGK^\dagger$ and $K^\dagger GK + E^\dagger GE$; the transfer $\mathcal{Z} = Z - EQ_0^+K + KQ_0^+E$; and the four maps α, γ that connect the middle-slot BPS states to the outer dressing.

Within this structure the monotone/fortuitous distinction is sharp and has a one-line origin, $[E, V] = K$. The tower is the only state whose dressing currents vanish by an identity, at every N. The other bare states are arithmetic walls that recede to peripheral sectors as N grows. Everything else is arithmetic too:

- which fortuitous states lift exactly through which channel (Laws 1–2);
- which are bare (eq. (6.2));
- when the tower's channel cost reaches CV's bound (eq. (7.2)).

These laws are numerical, but they hold without exception outside $N = 5$. Together they turn the structure of CV's empirical Figs. 6, 7 and 9 into consequences of BPS dimensions and of the ranks of Q_0, E and K.

11.2 Towards Stage 2: what a chord computation must produce

[interpretation] Section 8 identifies the objects. One flavor needed one two-point function, of the pair operator $E = \eta\wedge$. Two flavors need three, of $\psi^\dagger\chi$, $\psi^\dagger\psi^\dagger$ and $\psi^\dagger$. They also need their *mixed* correlators, e.g. $\langle Eh, GKh'\rangle$, which carry the interference. The monotone/fortuitous distinction would then appear as follows:

- for the tower, a Ward identity making the mixed and diagonal terms cancel exactly;
- for fortuitous states, the generic, partially cancelling spectrum of the same quadratic form.

D7 computed the one-flavor BPS→continuum two-point function at fixed charge in closed form, an Al-Salam–Chihara measure. A two-flavor version needs chord rules for two flavors with a charge lattice (N_ψ, N_χ). We have not checked whether such rules exist in the literature; establishing that is the first task of Stage 2.

The closed forms of §9 turn this into a definite list, valid exactly and not only at first order. The exact uplift needs:

- the self-energies $\mathcal{S}_G$, $\mathcal{S}'_G$ (two-point functions of E and K through the old continuum);
- the transfer $\mathcal{Z}$ (the direct Z coupling plus the zig-zags EQ_0^+K, KQ_0^+E);
- the maps α, γ between middle-slot BPS states and the outer dressing.

A chord computation of these three ingredients would give the exact two-flavor uplift.

11.3 Open problems

1. The geometry of the singular set of $\mathbb{M}$ (pure-dressing states, §9.4). Does it approach unit coupling as N grows?
2. Derivations of Laws 1–2, i.e. why the decoders are maximally generic inside their rooms. This is the analogue of CV's generic-position hypothesis, now with the refined rooms (7.1).
3. The $N = 5$ coincidences.
4. Simplifying the decoder pencils of Proposition 5 further, e.g. by push-through identities as in the one-flavor case.
5. Stage 2: the chord computation of $\mathcal{S}_G$, $\mathcal{S}'_G$, $\mathcal{Z}$, α and γ.

A Verification summary

Table 9: Checks of the exact statements (`tests/test_two_flavor.py`, 16 tests, and the production data). Residuals are maxima.

Statement	Sizes	Residual
$Q^2 = 0$; $[Q, V] = 0$ (symmetrized only); tower BPS, CV norms (86)	$N = 4, 5$	10^{-12}
Block form (2.4) as an operator identity	$N = 3$, both models	10^{-12}
$[Q_0, E] = [Q_0, K] = 0$, $[E, K] = \{Q_0, Z\}$; $[E, V] = K$ (symmetrized)	$N = 4$, both models	10^{-12}
Slot equations (Table 2) on all BPS states	$N = 4$, both models	10^{-10}
Theorem 1 (step 1): admissible solve; obstructed residual; splitting	$N = 4, 5$, both models	10^{-12}; ≥ 1; 10^{-13}
Two-step dimension formula (4.2)	$N = 2$–7	exact counts
Four-slot injectivity; reconstruction from harmonic parts	$N = 2$–7	deficit 0; 10^{-14}
Tower bare, slot weights	$N = 2$–7	4×10^{-16}
Single-slot first order = two-point functions (8.2)	$N = 4$	10^{-8} (relative)
Tower dark at first order	$N = 5$	10^{-10}
Closed forms (added 25 October)		
$\dim \mathsf{B}_{N+1} = \dim H(E_1, D)$ (Theorem 2)	$N = 2$–7, both models	exact counts
$(1 + \mathcal{K})^{-1}$ reconstructs BPS states; $\dim \mathcal{A} = \dim \mathsf{B}_{N+1}$; BPS residual	$N = 2$–7, both models	2×10^{-12}; exact; 10^{-12}
Self-energy form of A, A'; $L' + L^\dagger = C$, factorized leak	$N = 3$–6, both models	10^{-13}
Admissibility (a)–(c) = Theorem 3; decoder pencils	$N = 3$–6 (pencils $N \leq 7$)	exact; 2×10^{-11}
$\mathrm{Herm}\,\mathbb{M} \geq 1 - \|C\|/2$	$N = 2$–7, both models	no violation

B Numerical methods and reproducibility

- Couplings: `rng = default_rng(seed)`, with G i.i.d. complex Gaussian, $\mathbb{E}|G|^2 = 2$.
- Symmetrized couplings $C = (G + G^{T_{jk}})/2$: function `random_symmetric_couplings`.

- Generic couplings $C = G$: function `random_generic_couplings`.
- The old model is the restriction of the $(N+1)$-pair couplings to the first N pairs; the new pair sits at the top two bits of the Fock basis.
- All linear algebra is dense. BPS spaces are kernels of the Laplacian with relative tolerance 10^{-9}.
- Seeds: symmetrized 1–5 ($N = 2$–6) and 1–2 ($N = 7$); generic 1–3 ($N = 2$–6) and 1 ($N = 7$).
- All runs used `scripts/memwatch.py`; the largest peak among completed runs was 3.65 GB, at $N = 7$. A first seed-1 job at $N = 7$ cached all first-order pseudo-inverses; it was stopped at 4.6 GB, before writing any first-order records, and the first-order and tower tasks were rerun with per-sector cache release (`research/notes/two_flavor_uplift_2026-10-23.md`).

Item	Code and data
Model; uplift machinery	`src/two_flavor.py`, `src/two_flavor_uplift.py`; tests `tests/test_two_flavor.py`
Production runs	`scripts/run_two_flavor_uplift.py` (tasks `dims cv bare first tower e2 closed`) → `results/data/two_flavor_2026-10-23/*.jsonl`, logs, metadata
Pure-dressing states	`scripts/find_pure_dressing_states.py` → `pure_dressing_*.json`
Analysis, tables, figures	`scripts/analyze_two_flavor_uplift.py` → `summary.json`, `results/figures/two_flavor_2026-10-23/`
Derivations; research note	`docs/derivations.md` D8; `research/notes/two_flavor_uplift_2026-10-23.md`

References

[1] J. Chryssanthacopoulos and D. Vegh, *Fortuity and fragility in supersymmetric SYK*, arXiv:2608.12160 (2026).

[2] C.-M. Chang, Y. Chen, B. S. Sia and Z. Yang, *Fortuity in SYK models*, arXiv:2412.06902.

[3] M. Heydeman, G. J. Turiaci and W. Zhao, *Phases of $\mathcal{N} = 2$ Sachdev-Ye-Kitaev models*, arXiv:2206.14900.

[4] `syk_gravity` project, *Progress report, 13 October 2026: the uplift map and the origin of localization*, `research/pdfs/progress_2026-10-13.pdf`.

[5] `syk_gravity` project, *Progress report, 14 October 2026: the dressing spectrum from chords*, `research/pdfs/progress_2026-10-14.pdf`.